

Companion LXVIII: Cross-Correlation Functions of Persistence Diagrams in the Relaxation-Driven Cyclic Cosmology

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April 2026

Abstract

Cross-correlation functions of persistence diagrams quantify the coupling of topological features across different filtration parameters, smoothing scales, or redshift slices. In weak-lensing analyses, these correlations measure how the birth and death of connected components, loops, and void-like structures co-evolve across the gravitational field. In the standard cosmological model, cross-correlation functions reflect the nonlinear matter distribution and the projection of large-scale structure. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the filtration-coupled topology in a characteristic way. This Companion develops a continuous description of cross-correlation functions in RDCC, deriving how the relaxation scale influences topological coupling, persistence-lifetime alignment, and the observational signatures in weak-lensing surveys. The resulting framework provides a distinctive probe of the RDCC gravitational field through filtration-coupled topology.

1 Introduction

Persistence diagrams encode the birth and death of topological features in weak-lensing convergence fields. While cross-covariance measures the correlated spread of topological features across ensembles, cross-correlation functions quantify the coupling of topological structure across different filtration parameters, smoothing scales, or redshift slices.

In the standard cosmological model, cross-correlation functions reflect the nonlinear matter distribution and the projection of large-scale structure. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the filtration-coupled topology in a scale-dependent manner, producing a distinctive signature in cross-correlation analyses.

2 Cross-Correlation Functions of Persistence Diagrams

Given two filtrations with persistence diagrams $D(\nu_1)$ and $D(\nu_2)$, the cross-correlation function is defined as

$$\xi_D(\nu_1, \nu_2) = \frac{\langle W_p(D(\nu_1), D^*) W_p(D(\nu_2), D^*) \rangle}{\sqrt{\langle W_p(D(\nu_1), D^*)^2 \rangle \langle W_p(D(\nu_2), D^*)^2 \rangle}}. \quad (1)$$

This function measures the degree of filtration-coupled topology.

3 RDCC Modification of Persistence Diagrams

In RDCC, the convergence evolves as

$$\kappa_{\text{RDCC}}(\ell) = \kappa(\ell) \left[1 - \chi_{\kappa} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (2)$$

This modifies the birth and death thresholds of topological features:

$$b_{\text{RDCC}} = b [1 - \chi_b(\ell/\ell_{\text{rel}})^{\alpha}], \quad (3)$$

$$d_{\text{RDCC}} = d [1 - \chi_d(\ell/\ell_{\text{rel}})^{\alpha}]. \quad (4)$$

Thus the RDCC filtration diagrams become

$$D_{\text{RDCC}}(\nu) = \{(b_{\text{RDCC}}(\nu), d_{\text{RDCC}}(\nu))\}. \quad (5)$$

4 Cross-Correlation Functions in RDCC

The RDCC cross-correlation function is

$$\xi_{D,\text{RDCC}}(\nu_1, \nu_2) = \xi_D(\nu_1, \nu_2) \left[1 - \chi_{\xi} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (6)$$

This modification reflects the relaxation-driven change in filtration-coupled topology.

5 Filtration-Coupled Topological Signatures of RDCC

The relaxation mechanism enhances the coherence of large-scale modes. Cross-correlation functions therefore exhibit:

- enhanced correlation between large-scale filtration thresholds,
- suppressed correlation between small-scale thresholds,
- a characteristic turnover near ℓ_{rel} ,
- modified redshift evolution of filtration-coupled topology,
- altered alignment of persistence lifetimes across thresholds.

These signatures provide a direct probe of the RDCC gravitational field through filtration-coupled topology.

6 Conclusion

Cross-correlation functions of persistence diagrams in the Relaxation-Driven Cyclic Cosmology reflect the scale-dependent evolution of the gravitational potential and the nonlinear matter distribution. The relaxation mechanism modifies the filtration-coupled topology in a scale-dependent manner, producing distinctive signatures in weak-lensing surveys. These effects provide a direct observational window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of filtration-coupled topology, nonlinear structure, and the interplay between light and gravity in RDCC.

References

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